Shravan Chandra

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Vancouver, BC • UBC Master of Data Science • Expected Jun 2027

Part-time: Jan 2027 • Full-time: Jul 2027 • Eligible to work in Canada (study permit, up to 24 h/week; PGWP-eligible)

Summary

Machine Learning Engineer and UBC Master of Data Science candidate with nearly 5 years of software and ML engineering at Bosch, deploying production ML and GenAI systems. Built pipelines processing 80+ TB of telemetry, real-time anomaly detection, and multi-agent data assistants. IEEE-published researcher in sign language recognition.

Education

University of British Columbia — Master of Data Science, Vancouver, Canada — Aug 2026 – Jun 2027

PES University — BTech in Electrical & Electronics Engineering | CGPA: 8.5/10.0 — Aug 2017 – May 2021

Skills

Languages: Python, JavaScript, TypeScript, SQL, R

ML / DL: PyTorch, TensorFlow, TensorFlow.js, Scikit-Learn, Transformers, OpenCV, MediaPipe, ONNX

GenAI: LangChain, LangGraph, RAG Systems, Prompt Engineering, AI Agents, LLM APIs, Vector DBs (Chroma)

Data & Cloud: PySpark, Databricks, Azure ML, Pandas, Power BI, MLflow

Infra / Tools: Docker, FastAPI, PostgreSQL, Redis, Git, Kubernetes, Jupyter

Experience

Senior ML Engineer, Bosch Global Software Technologies — Bengaluru, IN — Jan 2025 – Jun 2026

- Built and deployed Valfred, a multi-agent data assistant, reducing routine lookups from 10–15 minutes to approximately 1–2 minutes for ∼30 beta users. Answered ∼81% of ∼500 internal test queries correctly within three attempts (65% first-pass).
- Implemented vector-based schema retrieval and SQL error recovery; deployed the assistant through MLflow serving and Unity Catalog on Databricks.
- Architected a distributed Databricks batch pipeline ingesting 80+ TB of high-frequency ECU sensor telemetry; cut per-partition processing time by 50%+ via PySpark query rewrites and broadcast join optimization.
- Deployed real-time autoencoder anomaly detection over 12-second sensor windows, using Databricks for batch preprocessing; shortened component validation cycles by 4 months, with estimated €300K+ in annualized savings from earlier failure detection.
- Served as the primary technical contact for 5+ ML initiatives across India, Germany, and China, guiding architecture and deployment decisions.
- Mentored 3 interns who transitioned to full-time ML roles; authored onboarding guides and infrastructure standards adopted team-wide.

ML Engineer, Bosch Global Software Technologies — Bengaluru, IN — Aug 2023 – Dec 2024

- Built a real-time predictive failure pipeline using Elastic Net regression on AUTOSAR DEM debounce signals; achieved 0.92 recall and surfaced early-warning alerts 30 seconds before ECU system errors.
- Implemented online distribution-shift detection (KL divergence + hypothesis testing) across 100+ ECUs, flagging sensor anomalies with 95% accuracy under continuous high-frequency data ingestion.
- Led TensorFlow 1.x → PyTorch migration across a multi-GPU Azure environment, cutting training time by 48% and unblocking distributed training at scale.

Software Developer, Bosch Global Software Technologies — Bengaluru, IN — Aug 2021 – Aug 2023

- Built sensor analytics platform with real-time visualization and time-series forecasting; automated data pipelines reducing manual processing time by 45% across 25+ engineers.
- Automated PCB pricing estimation, cutting average error against actual quoted prices by 30% versus the previous tools and processing time from 90 to 5 minutes.

Projects

ASL Guide — Full-stack Accessibility App, 2025–2026 github/ASL-App

- Built a two-agent LangGraph workflow for text-to-ASL translation, automating 11 ASL grammar rules and generating structured glosses for handshape, motion, and non-manual markers.
- Reduced per-request input tokens by approximately 73% through Gemini context caching; implemented Redis and in-memory LRU caching for response reuse.
- Built a 36-class TensorFlow.js fingerspelling classifier with 95.5% accuracy on a held-out, class-stratified split (not signer-independent). Ran inference entirely in-browser using MediaPipe, with measured latency below 100 ms.
- Deployed with Docker, PostgreSQL, and async SQLAlchemy; implemented rate limiting, security headers, and an admin dashboard for usage analytics and feedback triage.

Research & Publications

Signer-independent sign language recognition, each stage informing the next. GitHub

Stage 3: Neural Sign Language Translator (VQ-VAE + Conformer) 2026 — In Progress

- Developing a factorized VQ-VAE and Conformer sequence-to-sequence model for landmark-based sign language translation, with hybrid CTC/attention decoding. Full training and evaluation pending.
- Implemented hand-dominance normalization and coordinate preprocessing for signer-independent evaluation.

Stage 2: Graph-Based Recognition (ST-GCN) 2025

- Benchmarked a spatial-temporal graph network over 209 landmark joints; it underperformed the Stage 1 Conformer, motivating the move to learned discrete tokens in Stage 3.

Stage 1: LandmarkConformer — Google ASL Signs (250 classes) 2024

- Built a multi-stream Conformer over hand, pose and face landmarks (positions, multi-scale velocities, hand geometry) with an adversarial signer discriminator; **76.65% top-1** across 250 classes (94K samples) on a split by signer, so no validation signer is seen in training.

Dynamic Sign Language Translator — IEEE ICCAR 2022 Apr 2022

Shravan Chandra, Venkatarangan MJ, Jyothi TN 10.1109/ICCAR55106.2022.9782661

- Published lightweight sign recognition system (MediaPipe + XGBoost) achieving 90%+ accuracy across 15 ASL phrases at IEEE International Conference on Control, Automation and Robotics.